iGEM 2026: Concrete Modelling Project Proposal

Two-Pillar Framework for the MMP-9 / HNE Wound Biosensor

Eton / Dry Lab

ESM-2 substrate design · Latent Bayesian optimisation · Multi-scale biosensor physics

Project Overview

The modelling project consists of two interlocking pillars that together address both ends of the biosensor design problem.

Pillar 1 (ML/DL) answers: *which peptide sequence should we use?* A protein language model (ESM-2) is fine-tuned on experimental cleavage data and used as the backbone for a Bayesian optimisation loop that generates novel, selectivity-optimised substrate candidates outside the training distribution.

Pillar 2 (Physics) answers: *how will that peptide perform on the device?* A multi-scale model of the wound biosensor translates protease concentrations in wound fluid into a predicted electrochemical signal, via an ODE model of the wound fluid protease network and a diffusion-reaction model of the electrode surface.

The two pillars connect at a single point: the substrate sequence output by Pillar 1 is passed into Pillar 2 as the peptide whose surface kinetics are modelled. Together they produce a fully computational design rationale for the biosensor, grounded in experimental MEROPS data, that wet lab then validates.

Target panel (subject to wet lab confirmation): MMP-9 and HNE (Panel B)

Panel B is the recommended panel (see Panel B report): both models rated HIGH, both substrates fully supported by primary literature, no MMP-8/MMP-9 cross-reactivity issue to manage. If wet lab moves to Panel A (three-plex), both pillars extend trivially by adding a third protease head / ODE species.

iGEM Judging Criteria Alignment

The 2026 handbook lists four criteria for the Best Model award:

1. **How impressive is the modelling?** Both pillars use advanced methods uncommon in iGEM: protein language model fine-tuning, multi-objective Bayesian optimisation, ODE systems with parameter fitting, and Michaelis-Menten surface kinetics. The combination is novel.

2. **Did it help the team understand a part, device, or system?** Pillar 1 determines what to synthesise; Pillar 2 predicts how it will behave on the electrode. Neither is decorative.

3. **Were real measurements used?** Pillar 1 trains on MEROPS experimental cleavage data. Pillar 2 is parameterised from published wound fluid protease concentrations and literature kinetic constants. Both are validated against wet lab output.

4. **Good example for others?** Full pipeline (fine-tuned weights, BO configuration, ODE parameters) is documented and re-runnable on any protease pair.

The Heidelberg 2025 gold standard: model predicts optimal experimental condition, wet lab confirms it. Our equivalent: Pillar 1 outputs a novel substrate sequence; Pillar 2 predicts its LOD on the biosensor; wet lab measures both cleavage efficiency and device signal. The model made both predictions before the experiment ran.

Pillar 1: ESM-2 Fine-tuning and Latent Bayesian Optimisation

Stage 1a: Fine-tuning ESM-2 as a Selectivity Encoder

ESM-2 (`esm2_t33_650M_UR50D`, Lin et al. 2022) is pre-trained on 250 million protein sequences and provides rich evolutionary context as per-residue representations. The 650M variant is fine-tuned on the MEROPS MMP-9 and HNE training sets using two parallel classification heads:

$$h_k = \sigma\big(W_k \cdot \overline{\text{ESM2}}(x) + b_k\big), \quad k \in \{\text{MMP-9, HNE}\} \tag{1}$$

where $\overline{\text{ESM2}}(x) \in \mathbb{R}^{1280}$ is the mean-pooled final-layer representation of the 8-mer x. Fine-tuning proceeds in two phases: head-only for 10 epochs (trunk frozen), then full fine-tuning at a low trunk learning rate ($\eta = 10^{-5}$, discriminative per layer) to prevent catastrophic forgetting.

The fine-tuned embedding space is semantically structured by selectivity: sequences that share protease recognition properties cluster together. This is the latent space in which Stage 1b operates.

Interpretability output: integrated gradients applied to the fine-tuned model produce per-position attribution scores $\alpha_{k,p}$ for protease k at position p (P4 through P4'). If $\alpha_{\text{MMP-9, P1}'}$ is highest among all positions, this independently confirms the Kridel 2001 S1' pocket result. Positions where model attribution and biochemical literature disagree are flagged for point-mutant synthesis.

Stage 1b: Latent Bayesian Optimisation

A Gaussian process with Matérn-5/2 kernel is trained on the MEROPS training set to map ESM-2 embeddings $z \in \mathbb{R}^{1280}$ to the two-dimensional selectivity vector $[s_{\text{MMP-9}}, s_{\text{HNE}}]$. The GP posterior $p(s \mid z)$ provides both a prediction and a calibrated uncertainty, which the acquisition function exploits.

Multi-objective Bayesian optimisation using Expected Hypervolume Improvement (EHVI, implemented in BoTorch) searches the fine-tuned latent space for Pareto-optimal sequences: maximise $s_{\text{MMP-9}}$, minimise s_{HNE} (for the MMP-9 channel) and vice versa. The search runs for 50 BO iterations over a background pool of 500k 8-mer candidates embedded by ESM-2.

PSSM filter: before any candidate is passed to Pillar 2 or wet lab, it must satisfy $\text{PSSM}_{\text{target}} > 0$ and $\text{PSSM}_{\text{off-target}} < 2$. This applies the existing mechanistic scorer as a biochemical gate, bridging the new generative pipeline and the original substrate predictor.

Positive control: if BO rediscovers `IPRTLTAA` or `AAPVSGIG` from the training set, end-to-end pipeline consistency is confirmed before novel sequences are committed to synthesis.

Stage 1c: Deliverables

- Fine-tuned ESM-2 weights and AUROC benchmark vs. VHSE8 + LR baseline.
- Per-position attribution maps per protease (comparison to literature).
- Pareto front in selectivity space; top 3–5 novel candidate sequences.
- Positive control validation (rediscovery of known substrates).

Pillar 2: Multi-scale Biosensor Physics Model

Layer 2a: Wound Fluid Protease Network (ODE)

The wound fluid is not a simple solution of a single protease at a known concentration. MMP-9 exists primarily as a pro-enzyme (pro-MMP-9) and requires activation; HNE is inhibited by α_1-antitrypsin (A1AT) present in wound exudate. The effective (active, uninhibited) enzyme concentration is what the biosensor actually measures.

A system of ODEs models the wound fluid compartment:

$$\frac{d[\text{pro-MMP-9}]}{dt} = k_{\text{prod}} - k_{\text{act}}[\text{plasmin}][\text{pro-MMP-9}] - k_{\text{deg}}[\text{pro-MMP-9}] \tag{2}$$

$$\frac{d[\text{MMP-9}]}{dt} = k_{\text{act}}[\text{plasmin}][\text{pro-MMP-9}] - k_{\text{inh}}[\text{TIMP-1}][\text{MMP-9}] - k_{\text{deg}}[\text{MMP-9}] \tag{3}$$

$$\frac{d[\text{HNE}]}{dt} = k_{\text{rel}} - k_{\text{A1AT}}[\text{A1AT}][\text{HNE}] - k_{\text{deg}}[\text{HNE}] \tag{4}$$

Parameters are taken from published kinetic studies (Lijnen 2001 for plasmin-MMP-9 activation; Werb 1997 for TIMP-1 inhibition; Travis & Salvesen 1983 for A1AT-HNE). The ODE system is solved at steady state for three wound conditions (healing, chronic non-healing, infected) using published wound fluid protein concentration ranges (Trengove 1999, Ladwig 2002, Watt 2009).

Output: the predicted range of active [MMP-9] and [HNE] that the electrode surface actually encounters, for each wound state. This is the realistic input to Layer 2b, replacing the naive assumption that total and active enzyme concentrations are equal.

Layer 2b: Electrode Surface Kinetics

The peptide substrate from Pillar 1 is immobilised at the electrode surface at surface density Γ (mol/cm^2). When protease arrives by diffusion, cleavage follows Michaelis-Menten kinetics at the surface:

$$\frac{d\Gamma_{\text{cleaved}}}{dt} = \frac{k_{cat}\,[E]\,\Gamma}{K_m + \Gamma/\delta} \tag{5}$$

where $[E]$ is the free enzyme concentration from Layer 2a, δ is the effective diffusion layer thickness (estimated from Cottrell equation for the electrode geometry), and k_{cat}, K_m are taken from literature for the chosen substrate.

A 1D diffusion-reaction PDE for the enzyme concentration profile near the electrode surface is solved numerically (finite difference, 50-node spatial grid):

$$\frac{\partial[E]}{\partial t} = D_E \frac{\partial^2[E]}{\partial x^2} - \frac{k_{cat}\,[E]\,\Gamma}{K_m + [E]} \tag{6}$$

Sensitivity analysis: Γ and the linker length (which affects the pre-exponential electron transfer factor) are swept over physically realistic ranges to identify the surface density and linker length that maximise signal-to-noise at the clinically relevant protease concentrations identified in Layer 2a. This directly informs electrode fabrication.

Predicted LOD: the model outputs a predicted limit of detection for each wound state, expressed in nM of active enzyme. This is a concrete, falsifiable prediction for wet lab to test.

Layer 2c: Deliverables

- Steady-state active enzyme concentrations for three wound states (healing, chronic, infected) with uncertainty ranges from parameter variation.
- Optimal electrode surface density Γ^* and linker length recommendation.
- Predicted LOD (nM active enzyme) for MMP-9 and HNE channels.
- Sensitivity analysis figures showing how LOD depends on Γ and k_{cat}.

How the Two Pillars Connect

Pillar 1 output	Pillar 2 input
Novel substrate sequence (e.g. `IPRTLTAA` or BO-generated)	k_{cat}, K_m for chosen sequence used in Layer 2b
Selectivity score (MMP-9 vs. HNE)	Confirms the sequence is not a confound in the LOD model
PSSM score	Biochemical plausibility check before LOD prediction

The narrative is linear and checkable: Pillar 1 picks the peptide; Pillar 2 predicts what signal that peptide will produce on the device at wound fluid protease concentrations. Wet lab then measures both the cleavage efficiency of the peptide and the device signal. Two predictions, one experiment.

Model–Wet Lab Integration

The wet lab experimental plan (constructs pES-01 through pES-08) produces three categories of data that feed directly into the model. This section describes each integration point explicitly, because model-driven experimental design is the primary criterion that separates award-winning iGEM models from decorative ones.

Integration Point 1: Cross-reactivity Matrix and Condition Number κ

The wet lab runs every construct against every protease (MMP-9, HNE, LasB / thermolysin, TEV) and assembles a normalised cross-reactivity matrix. **The dry lab set $\kappa < 30$ as the separability threshold for this matrix.** This is a model-derived criterion: the model analysed the predicted selectivity profiles of the substrates and determined that a condition number below 30 indicates the two channels are sufficiently independent to be resolved in a multi-protease wound environment.

The wet lab experiment then measures the actual κ. If $\kappa < 30$, the model-driven substrate selection is confirmed. If $\kappa \geq 30$, the model re-runs with the empirical cross-reactivity values as hard constraints and proposes a revised substrate pair. This is a direct Heidelberg-style prediction-confirmation loop, using a named, quantitative criterion.

Integration Point 2: k_{cat}/K_m Feeds into Pillar 2

The enzyme titration protocol (substrate held below K_M, five enzyme concentrations spanning two log units, three replicates, initial rates below 20% conversion) produces apparent k_{cat}/K_m in $\text{M}^{-1}\text{s}^{-1}$ for each construct-protease pair. This is the primary kinetic parameter in the Pillar 2 electrode surface PDE.

The integration runs in two steps. First, the Pillar 2 model predicts LOD using literature k_{cat}/K_m values before wet lab runs. Second, once the enzyme titration data are in, the model re-runs with the measured values and produces an updated LOD prediction. The difference between the two runs quantifies how far literature values diverge from the actual constructs, which is itself a reportable finding. The final LOD prediction is therefore grounded in the team's own experimental kinetics, not literature alone.

Integration Point 3: Substrate Ranking Across pES-01 to pES-08

Pillar 1 ranks the substrate sequences in pES-01 through pES-08 by predicted MMP-9 selectivity and HNE selectivity before wet lab runs. The Ni-NTA depletion assay (load cleavage reaction onto nickel spin column, measure sfGFP fluorescence in flow-through vs. total) produces a quantitative cleavage percentage for each construct at each protease concentration. The rank order from the depletion assay is compared to the Pillar 1 predicted rank order. Spearman correlation between predicted and measured selectivity is reported as a model performance metric.

This is the most direct test of whether the ML model generalises beyond its training set: the pES constructs are actual synthesised sequences, not sequences drawn from MEROPS.

Sensor Architecture Note

The sensor constructs are FLAG-sfGFP-substrate-His_6 fusion proteins. Protease cleavage separates the FLAG and His_6 tags; the Ni-NTA depletion assay quantifies cleavage by measuring sfGFP fluorescence in the column flow-through. This is a fluorescence depletion assay, not a FRET assay. The distinction matters for Pillar 2: the observable is not an energy transfer efficiency but a retention fraction on the nickel column, which is a binary (cleaved / uncleaved) readout made quantitative by fluorescence intensity. The sandwich ELISA (anti-FLAG capture, anti-His and anti-GFP detection) provides orthogonal quantification on the best three or four constructs.

Summary

Component	Methods	Key output	Judging criterion
ESM-2 fine-tuning	Transfer learning, multi-task	Selectivity encoder	01, 03
Attention analysis	Integrated gradients	Position importance maps	02, 04
Latent BO	GP + EHVI, BoTorch	Novel substrate candidates	01, 02
PSSM filter	Log-odds (existing)	Biochemical gate	02, 03
Wound fluid ODE	Nonlinear ODE system	Active enzyme concentrations	01, 02, 03
Electrode kinetics PDE	1D diffusion-reaction	Predicted LOD, optimal Γ	01, 02
Cross-reactivity matrix	Ni-NTA depletion, all vs. all	$\kappa < 30$ confirmation	02, 03
Enzyme titration	Initial rate kinetics	Measured k_{cat}/K_m for Pillar 2	03
Substrate ranking	sfGFP depletion assay	Spearman corr. vs. Pillar 1 prediction	03

The project satisfies the Heidelberg 2025 gold standard across three named, quantitative model-experiment loops: the $\kappa < 30$ separability threshold (model criterion, wet lab measures it), the LOD prediction updated with measured k_{cat}/K_m (model predicts, wet lab parameterises it), and the substrate selectivity ranking (model ranks, depletion assay confirms it). The two-pillar structure gives judges a complete model story from molecular sequence design through to clinical signal output, with wet lab data at every integration point.